

Question_1:

Code:

```
1  /*
2  Question No: 1
3  Given two rooted trees, determine whether they are isomorphic.
4  Two trees are isomorphic if they have the same structure,
5  even if their node numbers are different.
6  Node 1 is the root of both trees.
7  */
8  #include <iostream>
9  #include <map>
10 #include <vector>
11 #include <algorithm>
12 using namespace std;
13
14 map<vector<int>,int> mp;
15 int id=1;
16 int dfs(int u,int p,vector<vector<int>>& g){
17     vector<int> v;
18     for(int x:g[u])
19         if(x!=p)
20             v.push_back(dfs(x,u,g));
21     sort(v.begin(),v.end());
22     if(!mp.count(v))
23         mp[v]=id++;
24     return mp[v];
25 }
26 int main(){
27     int t;
28     cin>>t;
29     while(t--){
30         int n;
31         cin>>n;
32         vector<vector<int>> a(n+1),b(n+1);
33         for(int i=0;i<n-1;i++){
34             int u,v;
35             cin>>u>>v;
36             a[u].push_back(v);
37             a[v].push_back(u);
38         }
39         for(int i=0;i<n-1;i++){
40             int u,v;
41             cin>>u>>v;
42             b[u].push_back(v);
43             b[v].push_back(u);
44         }
45         cout<<(dfs(1,0,a)==dfs(1,0,b)?"YES":"NO")<<"\n";
46     }
47 }
```

Output:

Test Case_1:

```
2
3
1 2
2 3
1 2
1 3
NO
3
1 2
2 3
1 2
2 3
YES
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
2
3
1 2
2 3
1 3
3 3
NO
3
1 2
2 3
1 3
3 3
NO
PS C:\Users\Prabin Yadav\
```